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Kinetics of Proton-Excited Ar–F_2 and Kr–F_2 Mixtures

CHUNG-HSUAN CHEN AND M. G. PAYNE

(Invited Paper)

Abstract—**Mixtures of Ar and F_2 and of Kr and F_2 were excited with pulsed proton beams. Time-resolved spectra as well as time-integrated spectra were employed to study the detailed mechanism of production and quenching processes of ArF*, KrF*, Ar_2F*, and Kr_2F*. The kinetics of ArF*, KrF*, and XeF* lasers are discussed briefly.**

I. Introduction

SINCE the reports of emission spectra of rare gas-halide excimers by Golde and Thrush [1], [2], and by Velazco and Setser [3], [4], kinetics studies and the development of rare gas-halide lasers have become some of the most important topics in quantum electronics. By electron beam pumping or discharge with fast excitation, laser emission has been demonstrated for XeF*, XeBr*, XeCl*, KrF*, KrCl*, and ArF* [5]-[15]. In the ArF* and KrF* cases energies per pulse in excess of 100 J [7] have been demonstrated. Also, kilohertz repetition rates have been achieved by electric discharge pumping of Ar-F_2 mixtures. The rare gas-halide lasers can also be used as pumps for dye lasers. The possibility of using rare gas-halide lasers to do isotope separation has been pursued extensively because of high laser efficiency and fast repetition rates.

Brau and Ewing [16] excited rare gas-halogen mixtures with intense electron beams and obtained emission spectra. An ionic model was used to estimate the transitions of inert gas monohalides, and the harpooning mechanism has been used to estimate the cross sections of monohalide formation between excited rare gas atoms and halogen molecules. There are two repulsive potential curves (i.e., $X^2\Sigma$ and $A^2\Pi$) for ground state rare gas monohalides. Three molecular states, $B^2\Sigma$, $C^2\Pi$, and $D^2\Pi$, are formed from positive rare gas ions and negative halogen ions. The strongest UV emission band corresponds to transitions of $B \rightarrow X$ and $D \rightarrow X$. Usually, the transition of $C \rightarrow A$ is weak. High resolution emission and absorption spectra were used to obtain upper state vibrational frequencies and dissociation energies of XeF, XeBr, XeCl, XeI, and KrF [17]-[21]. Dunning and Hay [22] reported *ab initio* calculations of the electronic potentials and radiative lifetimes of KrF which arise from $Kr(^1S) + F(^2P)$, $Kr^*(^3P) + F(^2P)$, and $Kr^+(^2P) + F^-(^1S)$. Goodman and Brus [23] determined the ground state potential well of XeF by isolating XeF in solid Ne and Ar.

Excited rare gas halides can be produced from either excitation transfer processes or from ion-ion recombination. Velazco *et al.* [24] measured the rates of quenching of metastable Ar, Kr, and Xe by F_2 using the low pressure (< 10 torr) discharge method. The quenching rate constants of resonant excited rare gas atoms or dimers by F_2 have not been reported. Also, the relative efficiencies with which various excited states of rare gas atoms or dimers react with F_2 to produce rare gas halides have not been determined previously. The rate of electron attachment by F_2 has been measured [25] as $\sim 10^{-9}$ $cm^3 \cdot s^{-1}$ at room temperature collision energies. The ion-ion recombination rate between the positive rare gas ions and the negative halogen atomic ions is calculated as $\sim 10^{-6}$ $cm^3 \cdot s^{-1}$ [26], [27]. The detailed ion kinetics have not been studied.

Recently, Nakano *et al.* [28] have observed that electron-beam excitations of Ar-F_2 mixtures and Kr-F_2 mixtures yield

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broad emission bands with peaks at 2900 Å and 4200 Å, respectively. They [28] determined that the 2900-Å band arises from Ar_2F^* and the 4200-Å continuum from Kr_2F^*. Rokni *et al.* [29] measured the three-body quenching rate constant of ArF* by argon and suggested that the destruction of ArF* by argon may produce Ar_2F^*. Jacob *et al.* [30] and Rokni *et al.* [31] reported three-body quenching rates of KrF* by Ar and by Kr. However, there is no direct experimental evidence concerning the mechanism for producing Ar_2F^* and Kr_2F^*. Eden *et al.* [32] determined the radiative lifetime of KrF* as 6.8 ns. Quigley and Hughes [33], [34] reported the quenching rate constants of KrF* and Kr_2F^* by F_2. Wadt and Hay [35] have made an *ab initio* calculation of lifetimes for the six lowest states of Ar_2F.

In this work, time-resolved and time-integrated VUV- and UV-visible spectra of Ar-F_2 mixtures and Kr-F_2 mixtures excited by proton beams are used to obtain: two-body quenching rate constants of $Ar(^1P_1)$, $Ar(^3P_1)$, $Ar(^3P_2)$, $Ar_2^*(1_u)$, $Kr(^3P_1)$, $Kr(^1P_1)$, $Kr_2^*(1_u)$, ArF*, Ar_2F^*, KrF*, and Kr_2F^* by F_2; three-body quenching rate constants of ArF* by argon and KrF* by krypton; and radiative lifetimes of Ar_2F^* and Kr_2F^*. On the basis of experimental data, complete kinetic schemes are generated. The energy precurosrs of ArF*, Ar_2F^*, KrF*, and Kr_2F^* are determined. The mechanisms of ion-ion recombination to produce ArF* and KrF* are also discussed briefly.

II. Description of Apparatus

The schematic of the experiment is shown in Fig. 1. Pulses of 2-MeV protons are injected into the reaction cell to produce ion pairs and excited atomic states. In addition to direct excitation by the proton beam, low-level metastable and resonance states of rare gas atoms are produced through recombination and cascade processes. The 10-ns proton pulse can be repeated with repetition time intervals from 4 to 256 μs, or it can be triggered manually. The number of protons per pulse can be varied from $\sim 10^6$ to $\sim 10^7$. In this work the stability of proton beam intensity is controlled from pulse to pulse within 20 percent. The proton beam intensity was always set low enough to prevent the occurrence of superelastic collisions with electrons or other nonlinear effects due to collisions between excited states. The incoming pulse of protons triggers a time-to-pulse height converter, and the pulse is stopped by the detection of a photon at single photon detectors attached to VUV- and UV-visible spectrometers. The emitted VUV photons are resolved with a VUV monochromator (McPherson 235) set at a 4-Å resolution during the experiment and are detected by a Bendix channel electron multiplier. Near-ultraviolet and visible photons pass through a quartz window and enter a McPherson monochromator (model EU701) with a resolution of 4 Å. Photons within the bandpass of the monochromator are detected with a Bendix photomultiplier (model 7500). Both VUV- and UV-visible photons were detected in a time-resolved way using a previously described delayed coincidence counting technique [36], [37]. Data were stored in a Nuclear Data 4410 system and could be retrieved by X-Y plotting or by printing the number of counts in each channel on a teletype. By this means, time-resolved information was accumulated from many proton pulses and time histories of the populations were obtained without the necessity of large excited state populations.

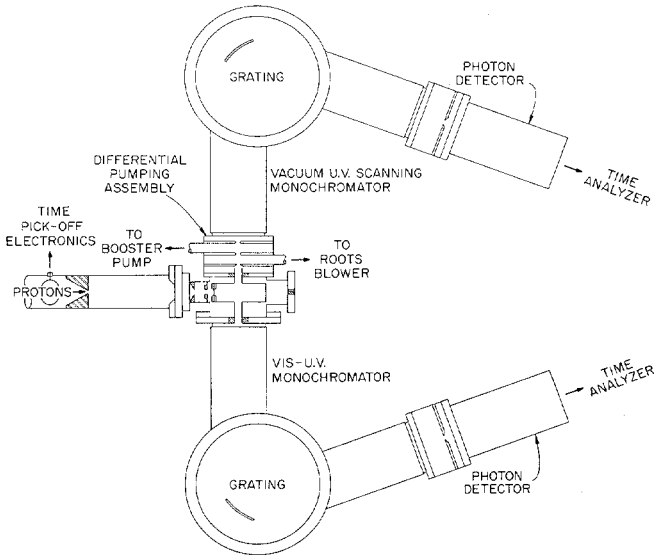


Fig. 1. Schematic of apparatus for time- and wavelength-resolved experiments.

The reaction cell was a stainless steel rectangular chamber 2 X 2 in on the front side and in length, 3.5 in. The protons entered the cell through a thin Havar foil (0.00025 cm) which covered a 6-mm aperture on the front side. Resonance and fluorescence radiation which escaped to the windows of the reaction chamber entered the dispersing VUV- and UV-visible monochromators, respectively, via a lithium fluoride window and a quartz window 0.714 cm in diameter.

Argon (99.9995 percent purity), Kr (99.995 percent purity), and F_2 (99.8 percent purity) at room temperature were allowed to flow through the reaction cell. Since fluorine gas is very reactive, 1 percent F_2 was passed through the reaction cell for about 20 h to passivate the cell before data were taken. A mixture of 1 percent F_2 in a rare gas (Ar or Kr) and pure gas (Ar or Kr) were passed through separate well-calibrated Hastings flow meters to obtain various F_2 partial pressures. The total pressure in the reaction cell was measured by an MKS Baratron pressure gauge. The accuracy of pressure readings was about 0.01 torr.

III. Results and Analyses

A. Ar-F_2

An emission spectrum from 1000 to 5000 Å for Ar-F_2 mixtures excited by a proton beam is shown in Fig. 2. The peaks at 1933 Å and 2840 Å correspond to the emission from ArF* and Ar_2F^*, respectively. The broad continua with peaks at 1100 Å and 1270 Å are from $Ar_2^*(O_u)$ and $Ar_2^*(1_u)$, respectively. Time-resolved emission spectra of ArF*, $Ar(^3P_1)$, $Ar(^1P_1)$, and $N_2[(C^3\Pi_u)_{v=0} \rightarrow (B^3\Pi_g)_{v=0}]$ are shown in Fig. 3 for Ar-N_2-F_2 mixtures excited by a proton beam for an argon partial pressure (P_{Ar}) of 30 torr, an F_2 partial pressure (P_{F_2}) of 1×10^{-3} torr, and an N_2 partial pressure less than 3×10^{-4} torr. Since the decays of $Ar(^3P_1)$ and $Ar(^1P_1)$ are much faster than the decay of $Ar(^3P_2)$ for low Ar and F_2 partial pressures and since $Ar(^3P_2)$ has been identified as the precursor of $N_2(C^3\Pi_u)_{v=0}$ [38], the major precursor of ArF*

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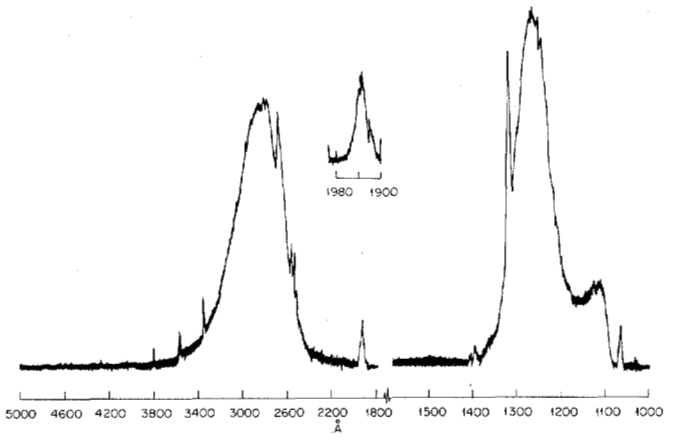


Fig. 2. Emission spectra of Ar-F_2 mixtures excited by a pulsed proton beam. Argon pressure is 413 torr and fluorine pressure is 0.055 torr. The average proton beam current is 0.9 μA, and the time interval between the proton pulses is 4 μs.

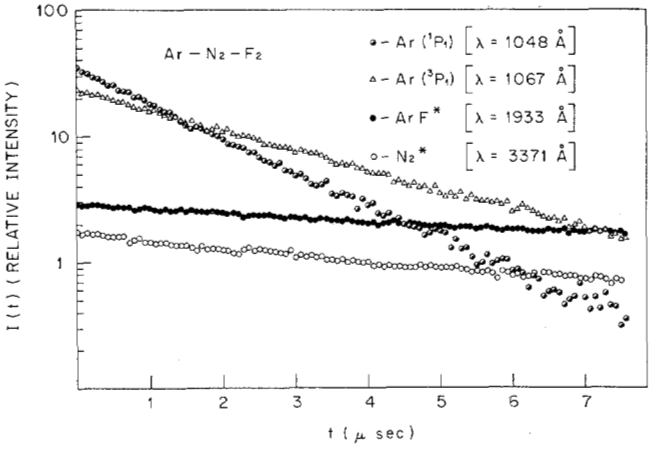


Fig. 3. Time-resolved emission spectra of Ar($^1P_1$), Ar($^3P_1$), ArF*, and N_2^* for Ar-N_2-F_2 mixtures excited by proton beams. The average proton beam intensity is 0.1 μA with a time interval between pulses of 8 μs.

can be identified as Ar($^3P_2$) for low argon pressures because of the similarity of the decay rates of ArF* and $N_2(C^3\Pi_u)_{v=0}$. For low argon pressure ($P_{Ar} < 100$ torr), no emission could be observed at 2900 Å.

The quenching rate constants for Ar($^3P_1$), Ar($^1P_1$), and $Ar_2^*(1_u)$ by F_2 are 8.9×10^{-10}, 13×10^{-10}, and 5.2×10^{-10} $cm^3 \cdot s^{-1}$, respectively, as shown in Figs. 4 and 5. In Fig. 6 the quenching rate constant of Ar($^3P_2$) by F_2 is obtained by monitoring the emission of N_2 $(C^3\Pi_u)_{v=0} \rightarrow N_2$ $(B^3\Pi_g)_{v=0}$. The quenching rate constant for Ar($^3P_2$) by F_2 is determined as 9.4×10^{-10} $cm^3 \cdot s^{-1}$ which agrees with the discharge results obtained by Velazco *et al.* [24]. The decay frequency of ArF* is also plotted in Fig. 6. It is confirmed that the precursor of ArF* at low argon pressure is Ar($^3P_2$).

Since Ar($^3P_1$) and Ar($^1P_1$) do not produce significant amounts of ArF* and since the population of Ar($^3P_0$) is much less than that of Ar($^3P_2$), only Ar($^3P_2$) and $Ar_2^*(1_u)$ are important for producing ArF*. Since the electron density is always kept low ($< 10^{12}$ cm^{-3}), the quenching of excited atoms or molecules by electron collisions can be neglected. Thus a unique and complete kinetic scheme for ArF* and Ar_2F^* produced from neutral excited argon atoms and dimers

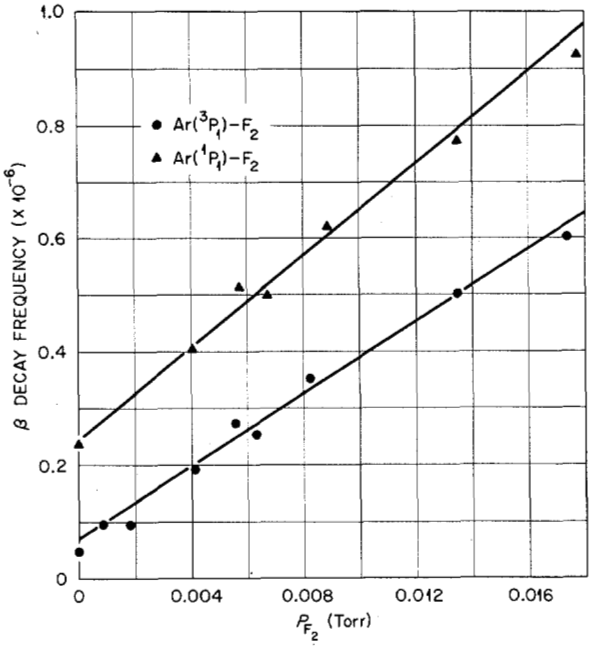


Fig. 4. Quenching rates of Ar($^3P_1$) and Ar($^1P_1$) by F_2. The argon pressure is 205 torr.

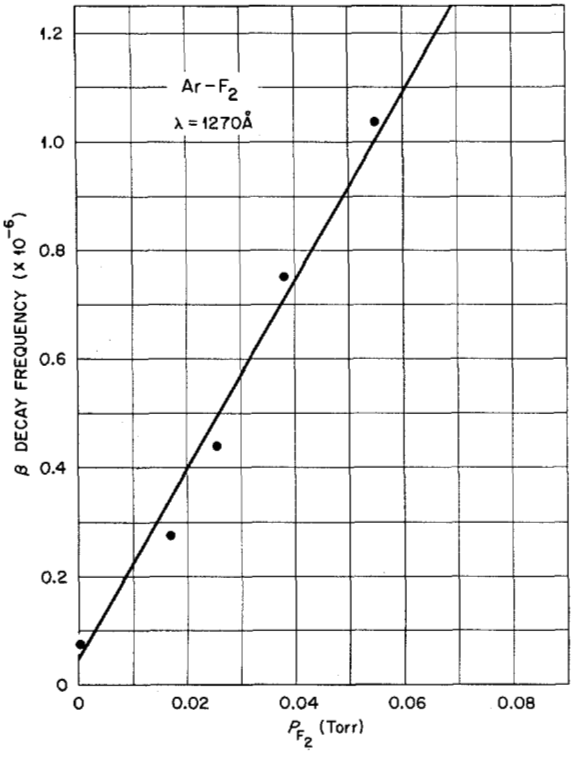


Fig. 5. Destruction rate of $Ar_2^*(1_u)$ by F_2. The argon pressure is 605 torr.

at high argon partial pressure in Ar-F_2 mixtures excited by low intensity fast charged particles is given in Fig. 7. The quenching processes of ArF* and Ar_2F^* by Ar through two-body collision are neglected since ArF* and Ar_2F^* have short lifetimes and the Ar pressure was kept less than 600 torr. The three-body quenching rates of Ar($^3P_2$) and ArF* by Ar are represented by β and β_3', respectively; β_1, β_2, β_3, and β_4 are rate constants for quenching of Ar($^3P_2$), $Ar_2^*(1_u)$, ArF*, and

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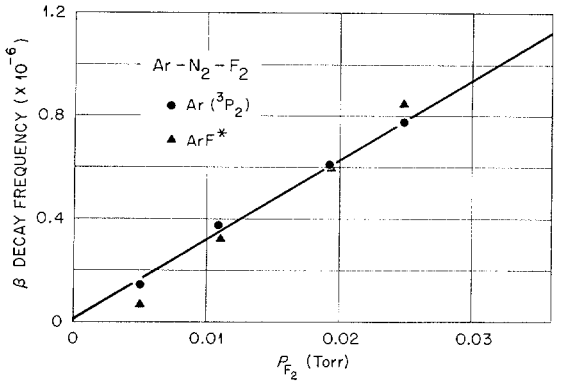


Fig. 6. Decay frequency of ArF* and $Ar(^3P_2)$ for various F_2 pressures with argon pressure at 36 torr. The N_2 partial pressure is estimated to be below 0.001 torr.

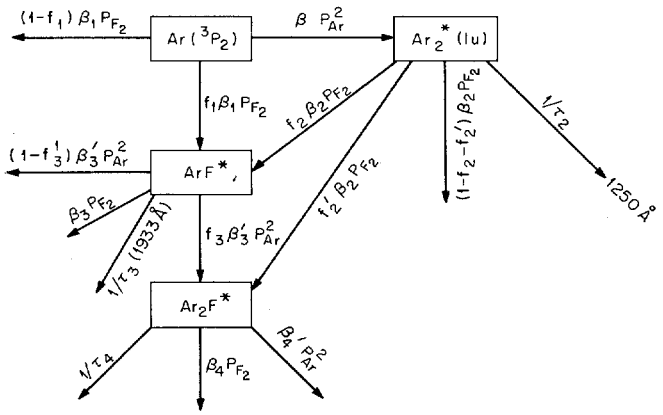


Fig. 7. Kinetic scheme for ArF* and Ar_2F^* produced by neutral excited argon atom and molecules.

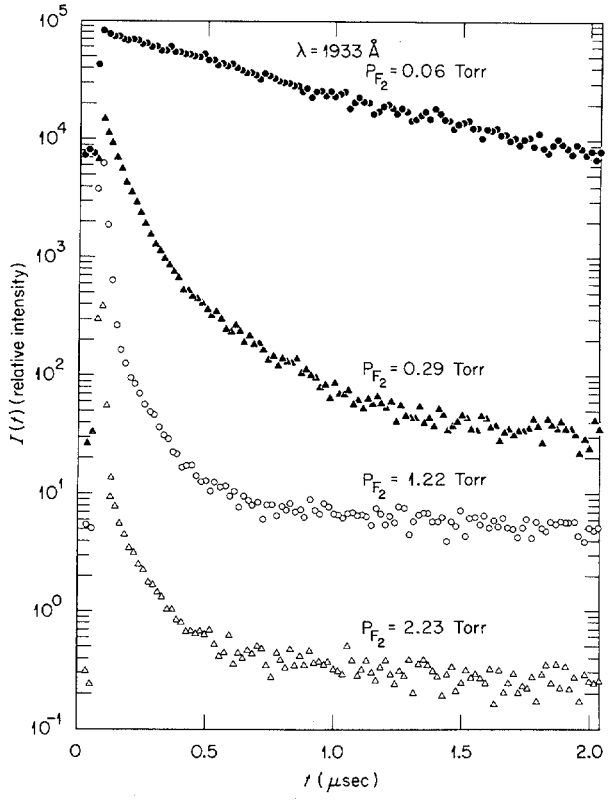


Fig. 8. Time-resolved emission spectra of ArF* with argon partial pressure at 600 torr.

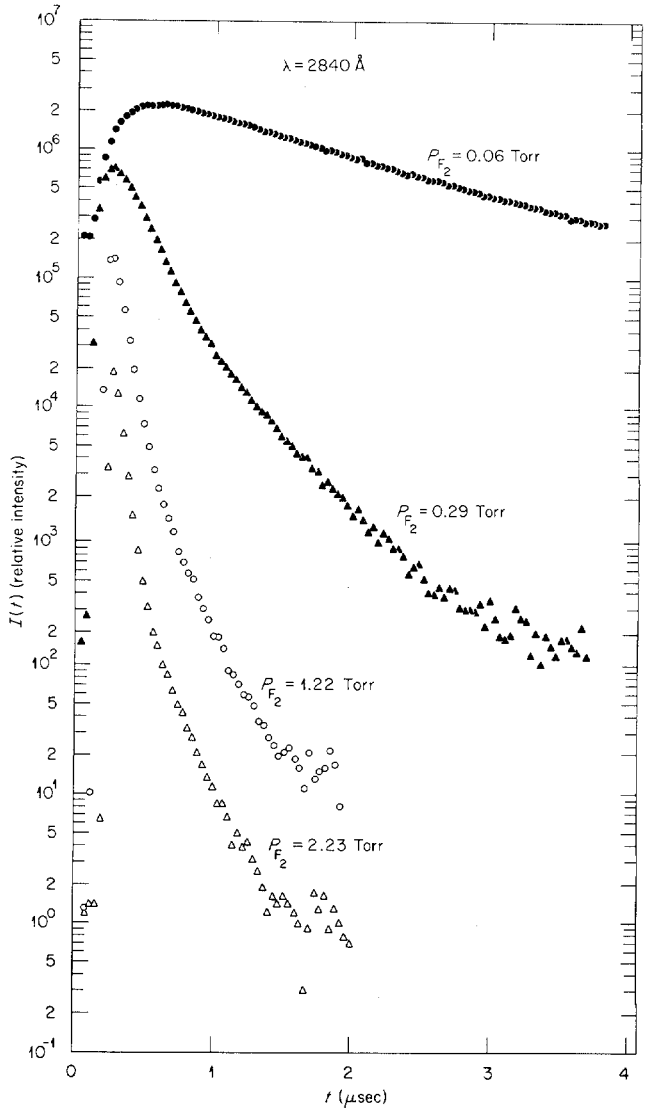


Fig. 9. Time-resolved emission spectra of Ar_2F^* with argon partial pressure at 600 torr.

Ar_2F^* by F_2, respectively. The radiative lifetimes of $Ar_2^*(1_u)$, ArF*, and Ar_2F^* are represented in Fig. 7 by τ_2, τ_3, and τ_4, respectively; f_1, f_2, f_2', and f_3 are the percentages of $Ar(^3P_2)$ quenched by F_2 to produce ArF*, $Ar_2^*(1_u)$ quenched by F_2 to form ArF*, $Ar_2^*(1_u)$ quenched by F_2 to produce Ar_2F^*, and ArF* quenched by three-body collisions with Ar to form Ar_2F^*, respectively. Our values of β and τ_2 are 11 ± 2 torr^{-2} $\cdot s^{-1}$ and 3 μs, respectively [39]. A value of 4 ns was obtained for τ_3 from an *ab initio* calculation [40]. In the following, the experimental data, combined with a simple mathematical analysis, are used to obtain the numerical values of β_4, β_3, β_2, and τ_4. The relative efficiencies for producing ArF* by $Ar(^3P_2)$ and $Ar_2^*(1_u)$, as well as the mechanism for producing Ar_2F^*, will be pursued.

Time-resolved data with various F_2 partial pressures for ArF* and Ar_2F^* are shown in Figs. 8 and 9, respectively. There are fast and slow decay components for both ArF* and Ar_2F^*, and a long rise time (~200 ns) is observed for Ar_2F^* but not for ArF*. The slow decay parts are probably from ion-ion recombination between Ar_2^+ and F^-. A more detailed discussion will be presented later. In Fig. 10 the decays of ArF* and Ar_2F^* are very similar for low F_2 pressure. On the other

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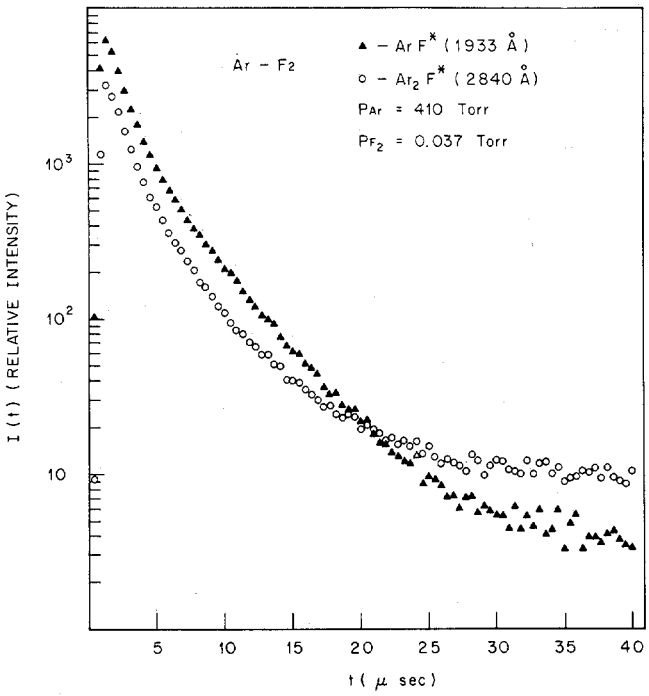


Fig. 10. Time-resolved spectra of ArF* and Ar_2F^*.

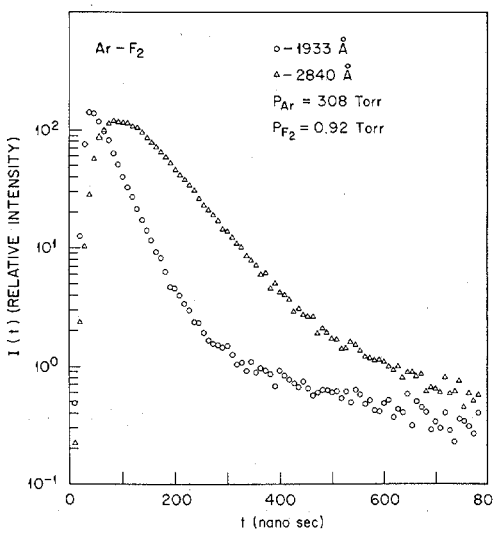


Fig. 11. Time-resolved emission spectra of ArF* and Ar_2F^*.

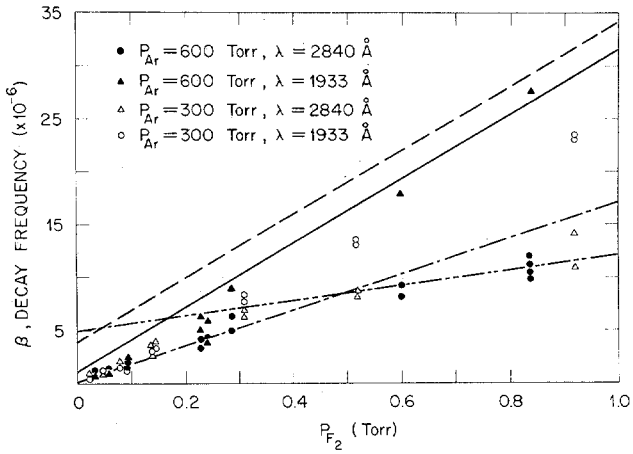


Fig. 12. Decay frequencies of ArF* and Ar_2F^* for various F_2 pressures. —·——, ---, represent the calculated decay frequencies of $Ar_2^*(1_u)$, $Ar(^3P_2)$, (P_{Ar} = 300 torr), and $Ar(^3P_2)$ (P_{Ar} = 600 torr). The numerical values of the quenching rates of $Ar(^3P_2)$ and $Ar_2^*(1_u)$ by F_2 are obtained from the present work.

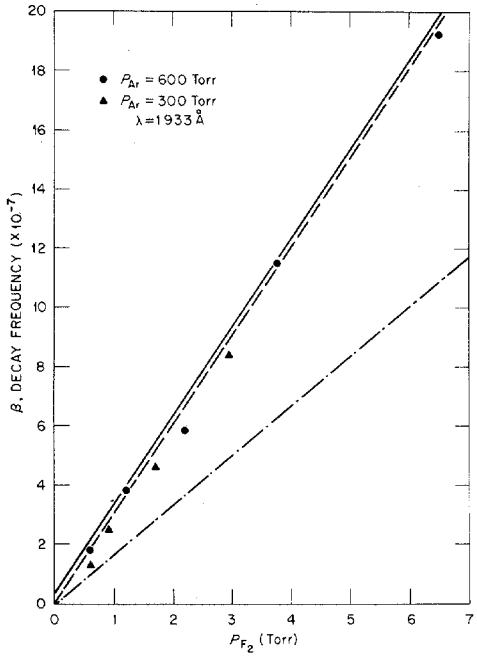


Fig. 13. Decay frequency of ArF* for various F_2 pressures. —·—, ---, and —— represent the calculated decay frequencies of $Ar_2^*(1_u)$, $Ar(^3P_2)$ (P_{Ar} = 300 torr), and $Ar(^3P_2)$ (P_{Ar} = 600 torr).

hand, the decay of Ar_2F^* is much slower than that of ArF* for high F_2 partial pressure (cf., Fig. 11). The decay frequencies of ArF* and Ar_2F^* for fluorine partial pressure less than 1 torr are plotted in Fig. 12. The decay frequencies of both ArF* and Ar_2F^* are identical within the experimental accuracy for fluorine partial pressures less than 0.1 torr. In Fig. 12, the decays of both ArF* and Ar_2F^* almost follow the decay of $Ar_2^*(1u)$ for F_2 pressure less than 0.1 torr and argon pressure greater than 300 torr. On the other hand, the decay of Ar_2F^* is slower than the decay of ArF* when F_2 partial pressure is higher than 0.5 torr. Decay frequencies of ArF* for various high F_2 pressures ($P_{F_2} > 0.5$ torr) are plotted in Fig. 13. The decay of ArF* follows the decay of $Ar(^3P_2)$ within experimental accuracy. This agrees with the previous conclusion that $Ar(^3P_2)$ is the major precursor for ArF* when F_2 partial pressure is high. The decay frequencies of Ar_2F^* for high F_2 partial pressure ($P_{F_2} > 1$ torr) are given in Fig. 14. It is found that the decay frequency of Ar_2F^* is much slower than the decay of $Ar(^3P_2)$ and $Ar_2^*(1_u)$. Thus we conclude that the decay of Ar_2F^* for high F_2 pressure ($P_{F_2} > 1$ torr) reflects the lifetime of Ar_2F^* (including Ar_2F^* quenched by F_2) instead of the decay of the precursors of Ar_2F^*. The above conclusion can be obtained by solving the following differential equations which are written according to the scheme shown in Fig. 7:

$$\frac{dN_1}{dt} = -(\beta P_{Ar}^2 + \beta_1 P_{F_2}) N_1, \tag{1}$$

$$\frac{dN_2}{dt} = \beta P_{Ar}^2 N_1 - \left(\beta_2 P_{F_2} + \frac{1}{T_2}\right) N_2, \tag{2}$$

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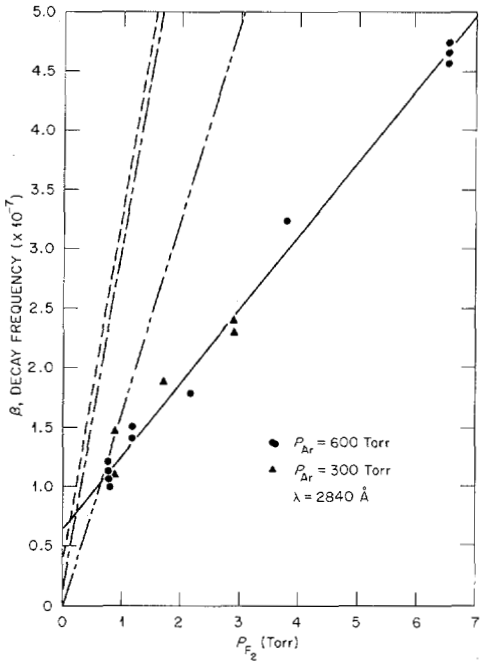


Fig. 14. Decay frequency of Ar_2F^* for various F_2 pressures. —·—·—, —·—, and – – – represent, respectively, the calculated decay frequencies for $Ar_2^*(1_u)$, $Ar(^3P_2)$ (P_{Ar} = 300 torr), and $Ar(^3P_2)$ (P_{Ar} = 600 torr).

$$\frac{dN_3}{dt} = f_1\beta_1 P_{F_2} N_1 + f_2\beta_2 P_{F_2} N_2 - \left(\beta_3 P_{F_2} + \beta_3' P_{Ar}^2 + \frac{1}{\tau_3}\right) N_3, \quad (3)$$

$$\frac{dN_4}{dt} = f_3\beta_3' P_{Ar}^2 N_3 + f_2'\beta_2 P_{F_2} N_2 - \left(\beta_4 P_{F_2} + \beta_4' P_{Ar}^2 + \frac{1}{\tau_4}\right) N_4. \quad (4)$$

In the above equations, N_1, N_2, N_3, and N_4 represent the populations of $Ar(^3P_2)$, $Ar_2^*(1_u)$, ArF*, and Ar_2F^*, respectively; P_{Ar} and P_{F_2} are the argon partial pressure and the fluorine partial pressure, respectively.

By solving the above differential equations, the populations of $Ar(^3P_2)$, $Ar_2^*(1_u)$, ArF*, and Ar_2F^* as a function of time can be obtained as follows:

$$N_1 = N_{10} \exp\left[-(\beta P_{Ar}^2 + \beta_1 P_{F_2})t\right] = N_{10} \exp(-\gamma_1 t), \quad (5)$$

$$N_2 = \frac{N_{10}\beta P_{Ar}^2}{\left(\frac{1}{\tau_2} + \beta_2 P_{F_2}\right) - (\beta P_{Ar}^2 + \beta_1 P_{F_2})} \cdot \left[\exp\left[-(\beta P_{Ar}^2 + \beta_1 P_{F_2})t\right] - \exp\left[-\left(\frac{1}{\tau_2} + \beta_2 P_{F_2}\right)t\right]\right] = \frac{N_{10}\beta P_{Ar}^2}{\gamma_2 - \gamma_1}(\exp(-\gamma_1 t) - \exp(-\gamma_2 t)). \quad (6)$$

Since the lifetime of ArF* is very short (~4 ns), the decay of N_3 can be obtained by taking $dN_3/dt = 0$.

$$N_3 = \frac{N_{10}P_{F_2}}{\beta_3' P_{Ar}^2 + \beta_3 P_{F_2} + \frac{1}{\tau_3}} \left[\left(f_1\beta_1 + \frac{f_2\beta_2\beta P_{Ar}^2}{\gamma_2 - \gamma_1}\right) \cdot \exp(-\gamma_1 t) + \frac{f_2\beta_2\beta P_{Ar}^2}{\gamma_1 - \gamma_2} \exp(-\gamma_2 t)\right] = I_1 \exp(-\gamma_1 t) + I_2 \exp(-\gamma_2 t), \quad (7)$$

$$N_4 = \frac{B_1}{\gamma_4 - \gamma_1} \exp(-\gamma_1 t) + \frac{B_2}{\gamma_4 - \gamma_2} \cdot \exp(-\gamma_2 t) - \left(\frac{B_1}{\gamma_4 - \gamma_1} + \frac{B_2}{\gamma_4 - \gamma_2}\right) \exp(-\gamma_4 t), \quad (8)$$

where N_{10}, the initial population of $Ar(^3P_2)$, depends on the gas pressure and proton beam intensity. The other parameters in the above equations are given as

$$\gamma_1 = \beta P_{Ar}^2 + \beta_1 P_{F_2}, \quad (9)$$

$$\gamma_2 = \beta_2 P_{F_2} + \frac{1}{\tau_2}, \quad (10)$$

$$\gamma_4 = \beta_4 P_{F_2} + \beta_4' P_{Ar}^2 + \frac{1}{\tau_4}, \quad (11)$$

$$I_1 = \frac{N_{10}P_{F_2}}{\beta_3' P_{Ar}^2 + \beta_3 P_{F_2} + \frac{1}{\tau_3}} \left(f_1\beta_1 + \frac{f_2\beta_2\beta P_{Ar}^2}{\gamma_2 - \gamma_1}\right), \quad (12)$$

$$I_2 = \frac{N_{10}P_{F_2} f_2\beta_2\beta P_{Ar}^2}{\left(\beta_3' P_{Ar}^2 + \beta_3 P_{F_2} + \frac{1}{\tau_3}\right)(\gamma_1 - \gamma_2)}, \quad (13)$$

$$B_1 = f_3\beta_3' P_{Ar}^2 I_1 + \frac{f_2'\beta_2 P_{F_2} N_{10}\beta P_{Ar}^2}{\gamma_2 - \gamma_1}, \quad (14)$$

$$B_2 = f_3\beta_3' P_{Ar}^2 I_2 - \frac{f_2'\beta_1 P_{F_2} N_{10}\beta P_{Ar}^2}{\gamma_2 - \gamma_1}. \quad (15)$$

From (8) we see that the decay part of Ar_2F^* represents the lifetime of Ar_2F^* when γ_4 is smaller than both γ_1 and γ_2. Thus the radiative lifetime can be obtained from Fig. 14 by extrapolation of the decay frequencies of Ar_2F^* to fluorine pressure equal to 0 torr. In this work, the three-body quenching of Ar_2F^* by argon is neglected since the decay frequency of Ar_2F^* is constant for argon partial pressures between 300 and 600 torr. From Fig. 14 the radiative lifetime of Ar_2F^* is 185 ns and the two-body quenching rate constant of Ar_2F^* by F_2 is 2.1×10^{-10} cm$^3 \cdot$s^{-1}. The time-integrated fluorescence intensity of Ar_2F^* can be expressed as

$$I_{Ar_2F^*} = \alpha\left(\frac{1/\tau_4}{\frac{1}{\tau_4} + \beta_4 P_{F_2}^2 + \beta_4' P_{Ar}^2}\right), \quad (16)$$

where α is a proportional constant. Thus,

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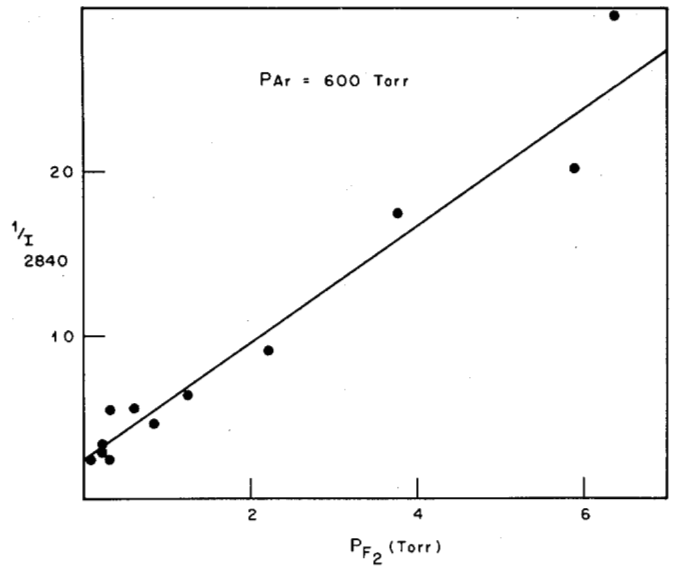


Fig. 15. Stern-Volmer plot for Ar_2F^* against F_2 pressure.

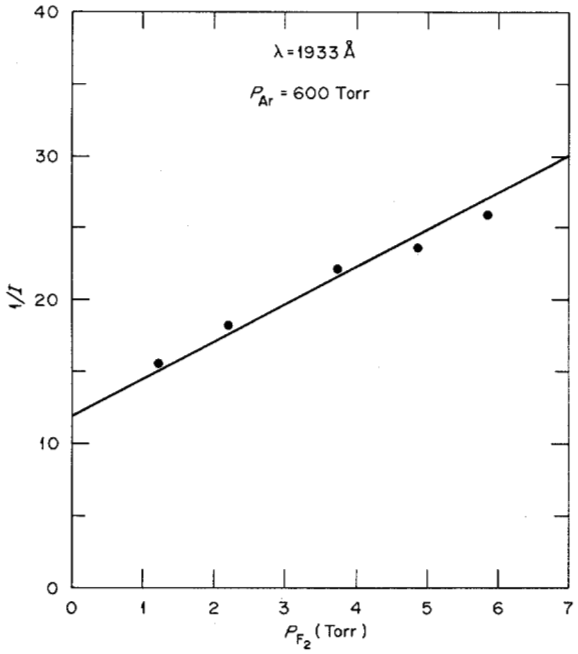


Fig. 16. Stern-Volmer plot for ArF* against F_2 pressure.

$$1/I_{Ar_2F^*} = \alpha'(1 + \tau_4\beta_4 P_{F_2}), \qquad (17)$$

$$\alpha' = 1/\alpha.$$

where

$$\beta_4' P_{Ar}^2 \ll \left(\frac{1}{\tau_4} + \beta_4 P_{F_2}\right).$$

The Stern-Volmer quenching curve for Ar_2F^* by F_2 is shown in Fig. 15. From Fig. 15, $\tau_4 \cdot \beta_4$ is determined to be 4.5×10^{-17} cm^3, in agreement with the results obtained from time-resolved spectra.

In Figs. 16 and 17, Stern-Volmer quenching curves are shown for ArF* with F_2 and with the square of argon pressure. Values of $\tau_3\beta_3$ and $\tau_3\beta_3'$ are obtained as 7.5×10^{-18} cm^3 and 2.1×10^{-39} cm^6, respectively. These results agree with the experimental data of Rokni *et al.* [29]. The lifetime of ArF* (τ_3) is calculated to be 4 ns [40]. Thus, β_3 and β_3' are 1.9×10^{-9} cm$^3 \cdot$ s^{-1} and 5.3×10^{-31} cm$^6 \cdot$ s^{-1}, respectively. All numerical values obtained in this work are tabulated in Table I.

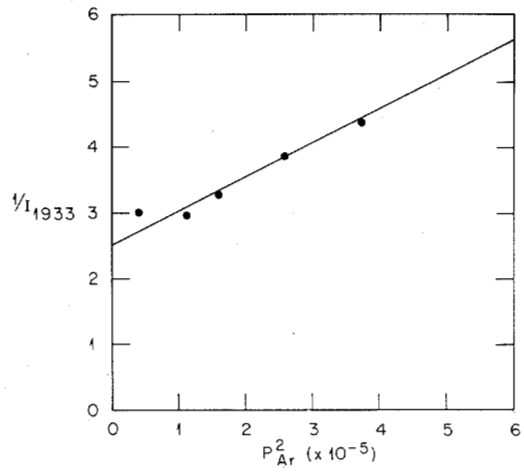


Fig. 17. Stern-Volmer plot for ArF* against P_{Ar}^2. P_{F_2} = 0.84 torr.

TABLE I
NUMERICAL VALUES FOR PARAMETERS IN THE KINETICS SCHEME OF Ar-F_2 MIXTURES SHOWN IN FIG. 7

$Ar(^3P_2) + 2Ar$	$\beta = 1.1 \times 10^{-32}$ cm^6 s^{-1}	[a]
$Ar(^3P_2) + F_2$	$\beta_1 = 9.4 \times 10^{-10}$ cm^3 s^{-1}	[b]
$Ar_2^*(1_u) + F_2$	$\beta_2 = 5.2 \times 10^{-10}$ cm^3 s^{-1}	[b]
$ArF^* + F_2$	$\beta_3 = 1.9 \times 10^{-9} \pm 25\%$ cm^3 s^{-1}	[c]
$ArF^* + 2Ar$	$\beta_3' = 5.3 \times 10^{-31} \pm 25\%$ cm^6 s^{-1}	[c]
Ar_2F^*	$\beta_4 = 2.1 \times 10^{-10} \pm 25\%$ cm^3 s^{-1}	[c]
$Ar_2^*(1_u)$	$\tau_2 = 3$ μs	[d]
ArF^*	$\tau_3 = 4$ ns	[e]
Ar_2F^*	$\tau_4 = 185 \pm 25\%$ ns	[c]
	$f_1/f_2 = 0.55 \pm 0.25$	[c]
	$f_2'/f_3 < 0.3$	[c]

[a] Reference [36].
[b] Reference [47].
[c] Present work.
[d] Reference [39].
[e] Reference [40].

From (7), the decay of ArF* is the sum of two exponential functions. If the difference between γ_2 and γ_1 is no more than a factor of 2, the early decay part (i.e., the first decay) can be represented as a single exponential function with a decay rate lying between γ_2 and γ_1. From time-resolved exponential data, N_3 can be written as a single exponential function of time for the decay part.

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$$N_3 \cong N_{30} \exp(-At) = (I_1 + I_2) \exp(-At). \tag{18}$$

Let

$$R = \frac{I_1}{I_2} = -1 + \frac{f_1}{f_2}\left(\frac{\beta_2}{\beta_1}\left(\frac{\gamma_1 - \gamma_2}{\beta P_{Ar}^2}\right)\right). \tag{19}$$

We choose A so as to match dN_3/dt as given by (18) and (7) at early times. Thus,

$$R = \frac{A - \gamma_2}{\gamma_1 - A}. \tag{20}$$

Substituting (20) into (19):

$$\frac{f_1}{f_2} = \frac{\beta_2 \beta P_{Ar}^2}{\beta_1(\gamma_1 - A)}. \tag{21}$$

The values of β_1, β_2, and β can be calculated and A can be obtained from the time-resolved emission of ArF*. Thus, f_1/f_2 can be determined. Experimental data of decay frequencies of ArF* for low F_2 pressure (<0.2 torr) are plotted in Fig. 18. The best fit of f_1/f_2 to the experimental results is determined to be 0.55 ± 0.25. For low F_2 pressure, the decay frequencies of Ar_2F^* are the same as the decay frequencies of ArF* (cf., Fig. 18). Thus, the second terms in (14) and (15) can be neglected. We conclude that f_2'/f_3 in Fig. 7 should be between 0 to 0.3 in order to agree with the experimental data. The destruction of ArF* from the three-body quenching process by argon should be the dominant channel for producing Ar_2F^*.

In Figs. 8 and 9 there are fast and slow decay parts for ArF* and Ar_2F^*. The fast decay part is due to energy transfer processes from neutral excited argon atoms or molecules. The slow decay part is considered to be from ion-ion recombination processes since the decay of the slow part is much slower than the decay of any excited argon atoms or dimers. Time-resolved emission spectra of Ar_2F^* excited by proton beams of different intensities are shown in Fig. 19. Rates of both the fast decay part and the slow decay part are the same for different proton beam intensities. The ratio of time-integrated intensity of the slow decay part to that of the fast decay part decreases when the proton beam intensity is reduced. The data shown in Fig. 19 also imply that the slow decay part is from ion-ion recombination processes.

The time histories of ArF* and Ar_2F^* from ion-ion recombination should follow second-order kinetics unless there is more than one kind of positive ion or negative ion. However, the experimental data in this work do not follow second-order kinetics for either ArF* or Ar_2F^*. This may imply that a second kind of negative ion can be formed in Ar-F_2 mixtures excited by proton beams. A possible kinetics scheme is shown in the following:

$$e + F_2 \longrightarrow F^- + F,$$

$$Ar^+ + 2Ar \longrightarrow Ar_2^+ + Ar,$$

$$F^- + F_2 + (Ar) \xrightarrow{\beta'} ?,$$

$$Ar_2^+ + F^- \longrightarrow ArF^* + Ar.$$

The differential equations for the above scheme can be written as

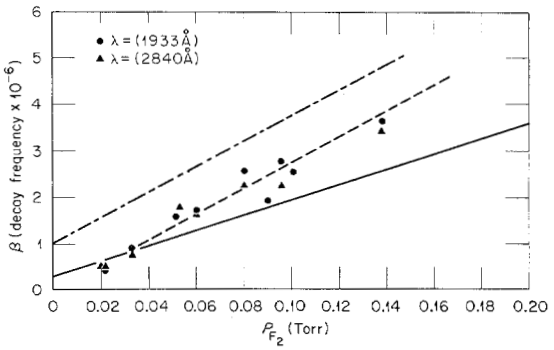


Fig. 18. Decay frequencies of ArF* and Ar_2F^* plotted for various F_2 pressures. ---·--- and —— represent the calculated decay frequencies of $Ar(^3P_2)$ and $Ar_2^*(1_u)$, respectively. ––– represents the decay frequencies of ArF* or Ar_2F^* when $f_1/f_2 = 0.55$. $P_{Ar} = 300$ torr.

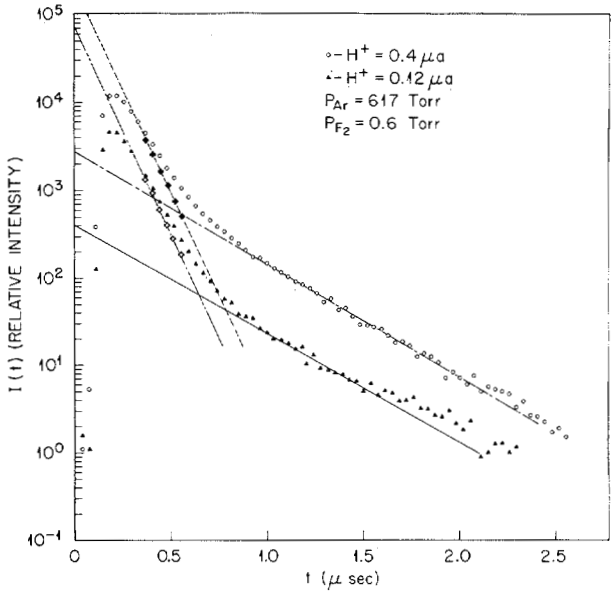


Fig. 19. Time-resolved spectra of Ar_2F^* for Ar-F_2 mixtures excited by various intensities of the proton beam. The time intervals between two proton pulses are 4 μs.

$$\frac{dN_+}{dt} = -\delta N_+ N_-,$$

$$\frac{dN_-}{dt} = -\delta N_+ N_- - \beta' P_{F_2} P_{Ar} N_-,$$

where N_+ and N_- represent the population of Ar_2^+ and F^-, respectively. Since the ion density in this work is low, the rate of ion-ion recombination is probably much slower than the quenching of F^-. Thus, the population of ArF* and Ar_2F^* should decay as

$$I = N_+ N_- = C \exp(-\beta' P_{F_2} P_{Ar} t),$$

where C is a proportionality constant.

The decay frequencies of the slow decay part of Ar_2F^* for various F_2 pressures with argon pressure at 600 torr are plotted in Fig. 20. From Fig. 20, the quenching rate of F^- by F_2 with argon pressure at 600 torr is obtained as 2×10^{-10} cm$^3 \cdot$ s^{-1}. Since collisions of F^- with F_2 for producing F_2^- at room temperature are not energetically possible, the quenching of F^- may need a third particle to stabilize products (such as F_3^-).

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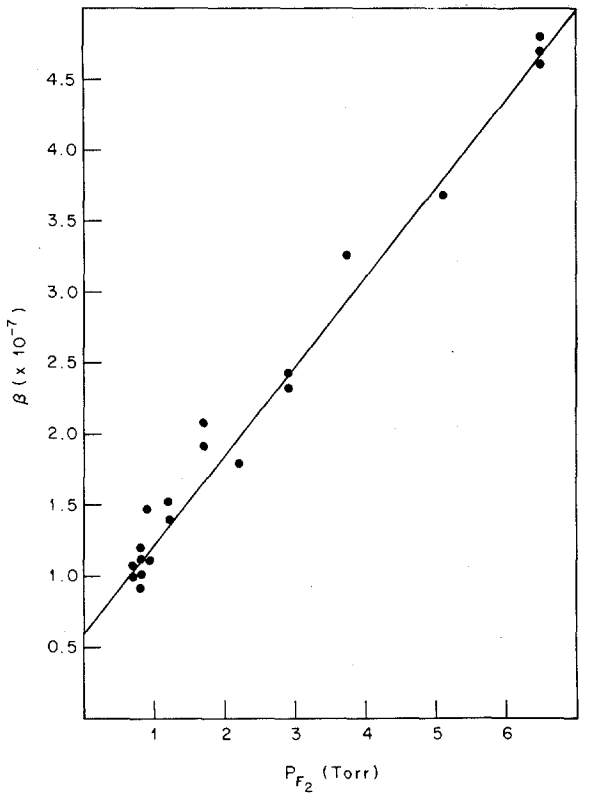


Fig. 20. Decay frequencies of F^- for various F_2 pressures with P_{Ar} = 600 torr.

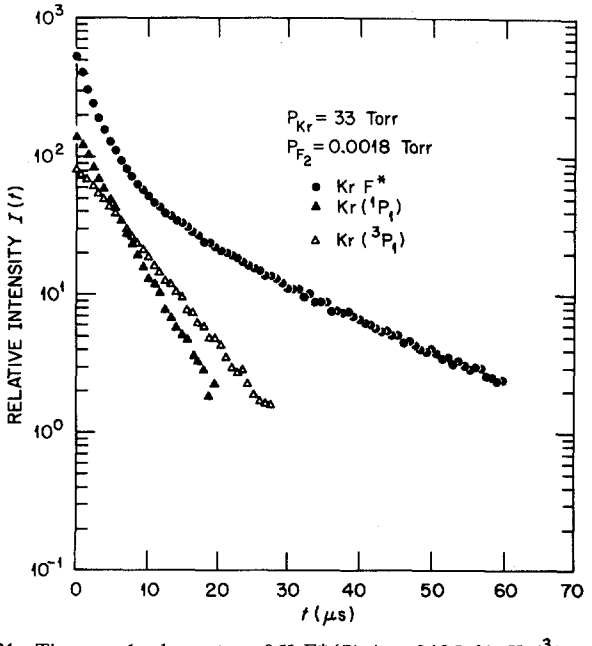


Fig. 21. Time-resolved spectra of KrF*(B) (λ = 2485 Å), Kr($^3P_1$) (λ = 1236 Å), and Kr($^1P_1$) (λ = 1165 Å). The average proton beam intensity is 0.06 μA. The time interval between two pulses is 64 μs.

More experimental work is definitely required to clarify the mechanism of the quenching of F^-.

B. Kr-F_2

Time-resolved fluorescence spectra of Kr($^1P_1$), Kr($^3P_1$) and KrF*(B) are shown in Fig. 21. There are two exponential decay constants for KrF*(B). Thus, KrF*(B) can be produced

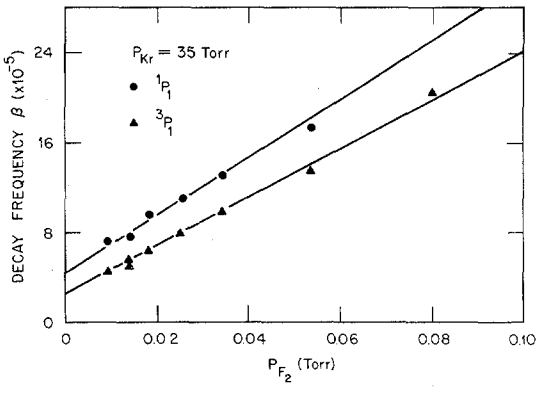


Fig. 22. Quenching rates of Kr($^3P_1$) and Kr($^1P_1$) by F_2.

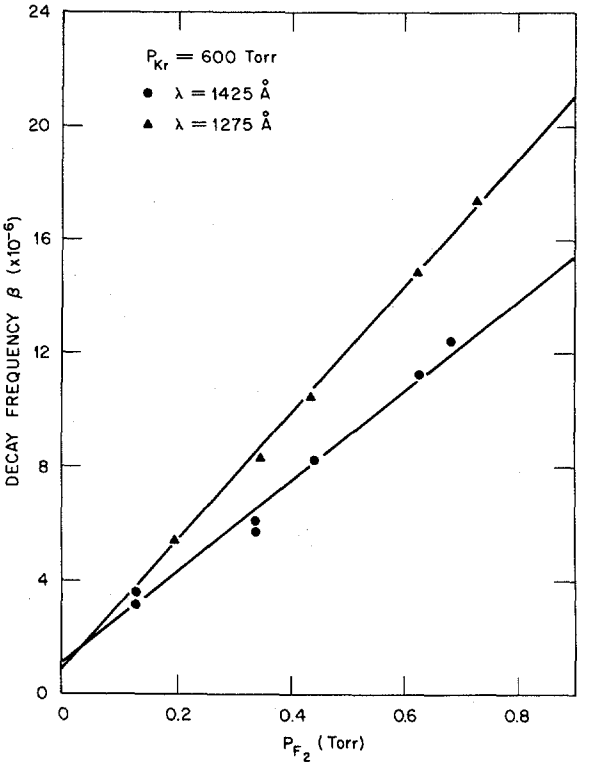


Fig. 23. Decay frequencies of $Kr_2^*(0_u)$ and $Kr_2^*(1_u)$ as determined from the 1275-Å continuum and the 1425-Å continuum, respectively. The lifetime of $Kr_2^*(1_u)$ is determined as ~1 μs. The lifetime of $Kr_2^*(0_u)$ is estimated as a few ns.

by resonance excited states of Kr as well as by metastable Kr. The population of KrF*(B) produced by resonance states of Kr is comparable to that produced by metastable Kr. The quenching rate constants of Kr($^3P_1$), Kr($^1P_1$), and $Kr_2^*(1_u)$ by F_2 are obtained from Figs. 22 and 23 as 6.8×10^{-10}, 7.6×10^{-10}, and 5×10^{-10} cm$^3 \cdot$ s^{-1}, respectively. The quenching rate constant of Kr($^3P_2$) by F_2 was measured as 7.2×10^{-10} cm$^3 \cdot$ s^{-1} by Velazco *et al.* [24].

Time-resolved spectra of KrF* and Kr_2F^* are shown in Fig. 24. There are fast and slow decay parts for both KrF* and Kr_2F^*, but the rise time of Kr_2F^* is much longer than that of KrF*. This indicates that the lifetime of Kr_2F^* is probably much longer than the lifetime of KrF*. A kinetic scheme for Kr-F_2 mixtures excited by a proton beam is shown in Fig. 25. The decay frequencies of Kr_2F^* for various F_2 pressures are

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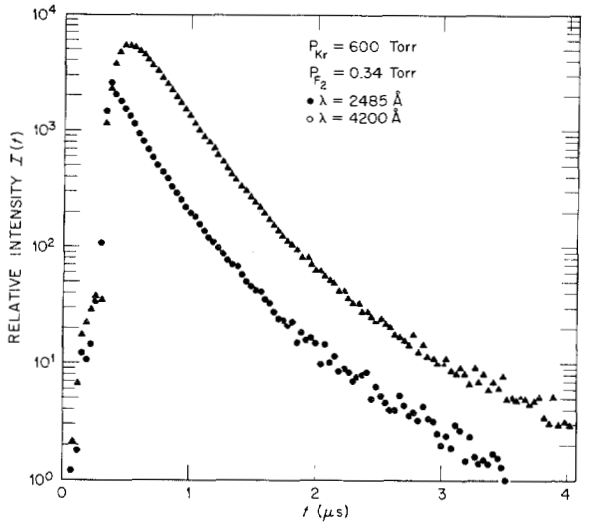


Fig. 24. Time-resolved emission spectra of KrF*(B) and Kr_2F^*. Since the wavelength resolution is 4 Å and the maximum transmission of the monochromator is set at 2485 Å, the emission spectra probably cover only B, not including D and C states.

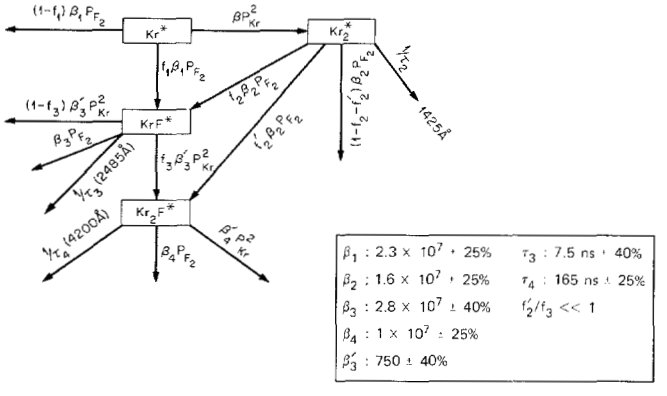


Fig. 25. Kinetic schemes for KrF* and Kr_2F^* produced by proton-excited Kr-F_2 mixtures.

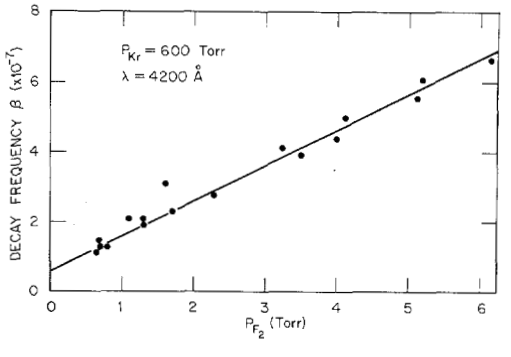


Fig. 26. Decay frequency of Kr_2F^* for various F_2 pressures.

plotted in Fig. 26. From an analysis similar to that already shown for Ar-F_2, the lifetime of Kr_2F^* is obtained as 165 ns, and the quenching rate of Kr_2F^* by F_2 is measured as 3×10^{-10} cm$^3 \cdot$ s^{-1}. The above results agree reasonably well with the experiments by Quigley and Hughes [34]. A Stern-Volmer plot of Kr_2F^* against F_2 pressures is shown in Fig. 27. We have determined that $\tau_{Kr_2F^*} \times k_4$ is equal to 5.6×10^{-17} cm^3, which agrees with the results from Fig. 26; k_4 is the quenching rate constant of Kr_2F^* by F_2. The reciprocals of KrF*(B) intensities are plotted against F_2 pressures in Fig. 28. A similar Stern-Volmer plot of KrF* for the square of Kr pressure is shown in Fig. 29. From Figs. 28 and 29 we can conclude that $\tau_{KrF^*} \times k_3$ and $\tau_{KrF^*} \times k_3'$ are equal to 6.6×10^{-18} cm^3 and 5.5×10^{-39} cm^6, respectively; k_3 and k_3' are quenching rate constants of Kr_2F^* by F_2 and 2Kr, respectively. From the rise time of KrF* emission, the lifetime of KrF* is estimated as 7.5 ns. Thus the quenching rate constants of KrF* by F_2 and by Kr through three-body collisions are 8.8×10^{-10} cm$^3 \cdot$ s^{-1} and 7.3×10^{-31} cm$^6 \cdot$ s^{-1}, respectively. The above results agree with the work by Jacob *et al.* [30], but the quenching rates are somewhat larger than those found by Eden *et al.* [32]. From time-resolved spectra of KrF*, Kr_2F^*, and $Kr_2^*(1_u)$, we conclude that the dominant process to produce Kr_2F^* is through KrF* as an intermediate species. The numerical values of quenching rates and lifetimes are tabulated in Fig. 25. The quenching rate constants shown in Fig. 25 are in units of torr $\cdot$ s^{-1}.

Time-resolved spectra of KrF*(B) excited by various proton beam intensities are shown in Fig. 30. The slow decay parts

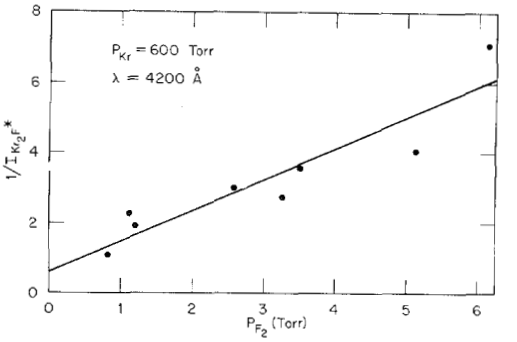


Fig. 27. Stern-Volmer plot for Kr_2F^* against F_2 pressure.

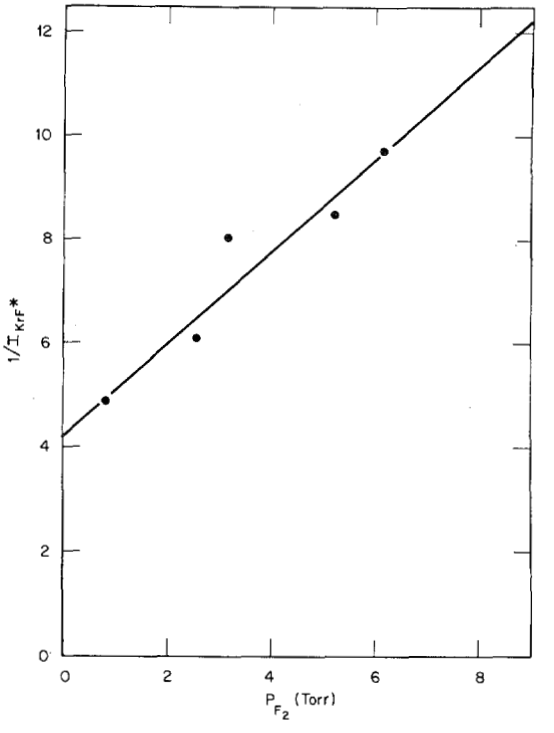


Fig. 28. Stern-Volmer plot for KrF*(B) against F_2 pressure. P_{Kr} = 600 torr.

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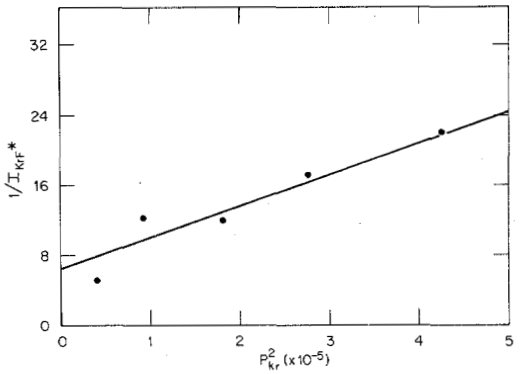


Fig. 29. Stern-Volmer plot for KrF*(*B*) against square of Kr pressure. The F_2 partial pressure is 0.66 torr.

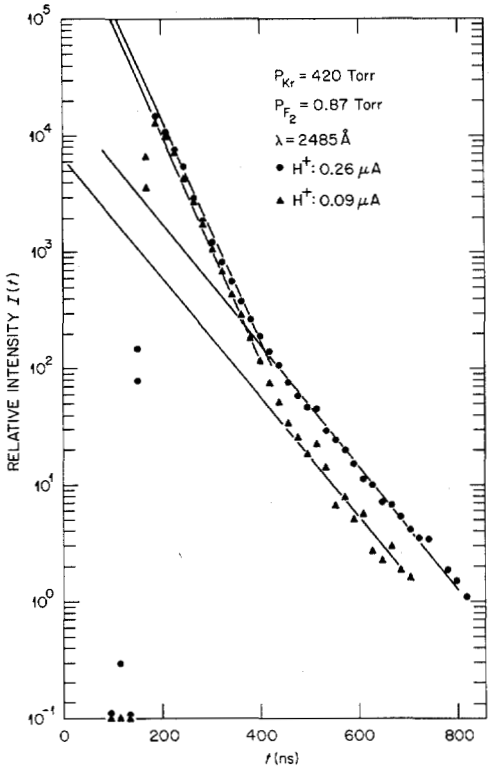


Fig. 30. Time-resolved spectra of KrF* for Kr-F_2 mixtures excited by various intensities of the proton beam.

are considered to be due to species produced in ion-ion recombination processes since the ratio of time-integrated intensity of the slow decay part to that of the fast decay part increases when the proton beam intensity increases. The decay frequencies of KrF* for various F_2 pressures are plotted in Fig. 31. The slope of the slow decay part is 2×10^{-10} $cm^3 \cdot s^{-1}$ which is about the same as the slope of the slow decay part of ArF* for Ar-F_2 mixtures excited by a proton beam. Thus, another indication is given that the quenching of F^- by F_2 plays an important role in the ion kinetics.

IV. Discussion

In this work we conclude that $Ar(^1P_1)$ and $Ar(^3P_1)$ do not produce ArF* very efficiently (<25 percent), although the quenching of $Ar(^1P_1)$ and $Ar(^3P_1)$ by F_2 is comparable to the quenching of $Ar(^3P_2)$ by F_2. On the other hand, $Kr(^3P_1)$ and $Kr(^1P_1)$ can produce KrF* with efficiency comparable to

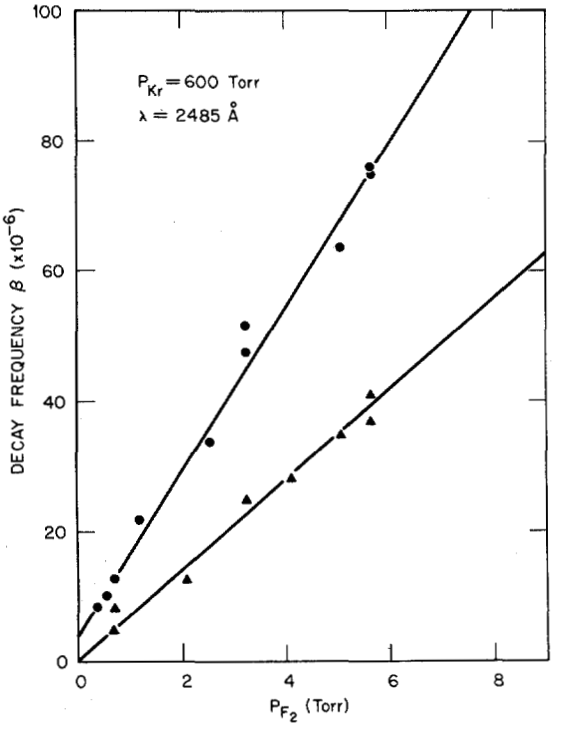


Fig. 31. Decay frequencies of KrF*(*B*) for various F_2 pressures. The fast decay part is represented by ·, while ▲ represents the slow decay part which is considered to be from ion-ion recombination.

$Kr(^3P_2)$. Krause *et al.* [41] showed in beam experiments that $Hg(6^3P_2)$ produces HgCl with a large cross section but that $Hg(6^3P_0)$ does not. The mechanism for such a highly selective quenching process has not been understood. The details of appropriate correlation diagrams are probably important in determining the final products. More studies are definitely necessary to understand this type of quenching mechanism. For *e*-beam pumped lasers or electric discharge lasers, various excited argon or krypton states are probably mixed by superelastic collision with electrons.

The three-body quenching rates of ArF* by Ar, KrF* by Kr, and XeF* by Xe are quite comparable. But it is observed that the production efficiency of Kr_2F^* is much less than that of Ar_2F^*. To our knowledge, evidence for the existence of Xe_2F does not exist. It should be interesting to discover the other reaction products. Recently, Loree and Brau [42] have observed that the fluorescence intensity of KrF* is increased by using Ne as a buffer gas instead of Ar. The detailed mechanism for the enhanced fluorescence requires further study.

Under our experimental conditions, the ratio of the intensity of ArF* or KrF* from ion-ion recombination to that from the energy transfer process is about 10 percent for F_2 pressures higher than a few torr, although the population of argon ions is always more than the population of excited argon atoms. If the intensity of fast charged particles is increased, the quenching of F^- will eventually become slower than the ion-ion recombination process and the ion-ion recombination rate constant can be measured. The intensity of ArF* and KrF*(*B*) might then be dominated by the ion-ion recombination process.

Since the lifetimes of rare gas halides have been measured and since the quenching rates of rare gas halide and F^- have

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been determined, the complete kinetics of rare gas halides can be uncovered if the absolute population of excited rare gases and the branching ratios for each reaction can be determined. Recently, Hurst *et al.* [43] and Payne *et al.* [44] used resonance ionization spectroscopy to determine the population of metastable He by proton excitation.

Resonance ionization techniques will be used to measure the population of various excited Kr and Ar species in the future. Since the proton beam can be well focused, the absolute fluorescence can be measured and used to determine the cross section of each individual branch reaction. In the past few years, the absorption spectra of KrF*, Kr_2F^*, and Ar_2^+ have also been extensively studied in order to estimate the relation between laser power and absorption coefficients.

Ewing and Brau [5] have used intense electron beams to demonstrate laser action of KrF* from mixtures of Ar, Kr, and F_2 in the ratio of 98.9:1:0.1 for a total pressure range of 1 to 5 atm. They [5] also achieved lasing of XeF* with mixtures of Ar, Xe, and F_2 in the ratio of 99.6:0.3:0.1. The quenching rate constants of $Ar(^3P_2)$ and $Ar(^3P_0)$ by Kr and Xe were measured by Piper *et al.* [45]; and Leichner [46] obtained the quenching rate constants of $Ar(^3P_1)$, $Ar(^1P_1)$, and $Ar_2^*(1_u)$ by Kr. Chen *et al.* [47] determined the quenching rate constants of $Ar(^3P_1)$, $Ar(^1P_1)$, and $Ar_2^*(1_u)$ by Xe. The quenching rate constants of excited Ar by Kr, Xe, and F_2 are tabulated in Table II. From the quenching rate constants of excited atomic and molecular argon by Kr, Xe, and F_2, the following mechanisms are proposed for the kinetics of the KrF* and XeF* lasers which were demonstrated by Ewing and Brau [5]:

$$Ar^* \text{ or } (Ar_2^*) + F_2 \rightarrow ArF^* + F + (Ar),$$

$$ArF^* + 2Ar \longrightarrow Ar_2F^* + Ar,$$

$$ArF^* + Kr \longrightarrow KrF^* + Ar.$$

For XeF* lasers, the mechanism becomes more complicated:

$$Ar^* \text{ or } (Ar_2^*) + Xe \rightarrow Xe^* + Ar \quad \text{or} \quad (2Ar),$$

$$Xe^* + F_2 \rightarrow XeF^* + F,$$

or

$$Ar^* \quad \text{or} \quad (Ar_2^*) + F_2 \rightarrow ArF^* + F + (Ar),$$

$$ArF^* + Xe \rightarrow XeF^* + Ar.$$

The quenching rate constants of ArF* by Kr and Xe have been determined as 1.6×10^{-9} and 4.5×10^{-9} $cm^3 \cdot s^{-1}$, respectively, by Rokni *et al.* [29], [31].

The ion-ion recombination processes and dissociative recombination processes should also populate KrF* and XeF* for *e*-beam pumped lasers. Since the charge transfer rates of Ar_2^+ to Kr and Xe are only 7.5×10^{-10} and 7.2×10^{-11} $cm^3 \cdot s^{-1}$, respectively [47], the mechanism for ion-ion recombination to populate KrF* and XeF* should go through ArF* as an intermediate state instead of recombination between Kr^+ or Xe^+ and F^- even when ion density is high ($\sim 10^{15}$ cm^{-3}).

TABLE II
QUENCHING RATE CONSTANTS AND QUENCHING CROSS SECTIONS OF $Ar(^3P_1)$, $Ar(^1P_1)$, $Ar(^3P_2)$, AND $Ar_2^*(1_u)$ BY F_2, Xe, Kr

	F_2	Xe	Kr
$Ar(^1P_1)$ k^a	12.9[b]	3.3[b] (3.47)[c]	0.012[c]
σ^d	225[b]	73[b] (77)[c]	0.25[c]
$Ar(^3P_1)$ k^a	8.9[b]	2.2[b] (1.92)[c]	0.091[c]
σ^d	156[b]	49[b] (43)[c]	2.3[c]
$Ar(^3P_2)$ k^a	9.4[b] (7.5)[e]	1.8[f]	0.062[f]
σ^d	165[b] (131)[e]	40[f]	1.3[f]
$Ar_2^*(1_u)$ k^a	5.2[b]	2.4[b] (1.92)[c]	0.091[c]
σ^d	105[b]	57[b] (46)[c]	1.9[c]

[a] The rate constant k has units of 10^{-10} $cm^3 \cdot s^{-1} \cdot molecule^{-1}$.
[b] Reference [47].
[c] Reference [46].
[d] $\sigma = kv = k/\bar{v}\ 8k'T/\tau\mu^{1/2}$, where k' is Boltzmann's constant and μ is reduced mass. σ has units of $Å^2$.
[e] Reference [24].
[f] Reference [45].

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